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ATTACHMENT J-1601000-01
DEFINITIONS AND ACRONYMS

Definition/Acronym	Description
Assessment	A general term referring to either a survey or inspection of a facility to determine condition.
Asset	A general term used to refer to an item, such as a component, system, building or facility, that is managed by an automated data management program, e.g., UICAP.
Business Process Management System (BPMS)	A web-based tool that provides a systematic method for the management of business processes, common practices, and process quality improvements that produce and support the most efficient and effective delivery of NAVFAC's products and services.
Customer Average Interruption Duration Index (CAIDI)	Customer average interruption duration in minutes per year. The electrical reliability benchmark is 60 minutes per year.
Facility Related Control System	Facility-related control systems are a subset of control systems that are used to monitor and control equipment and systems related to DoD real property facilities (e.g., building control systems, utility control systems, electronic security systems, and fire and life safety systems).
Industrial Control System	Most specifically a control system which controls an industrial process. Sometimes also used to refer to other types of control systems, particularly utility control systems such as electrical, gas, or water distribution systems. ICS includes PLCs, field sensors, field actuators, meters, transducers, relays, protection relays, contactors, electrical current, potential transformers, etc.
Integrated Maintenance Plan	IMP is a recurring state-of-the-art, reliability-centered inspection, testing, maintenance and repair program that determines best practices for managing the functions and consequences of failures of facilities equipment and system components. IMP encompasses accepted commercial practices, including reactive, preventive, predictive and proactive maintenance, into one optimal program. The IMP approach gives the Contractor full responsibility to maintain systems and equipment and perform repairs whenever necessary to ensure equipment and systems are operational and remain in a constant state of readiness.
In-house	Performed by Government employees.
Inspection	A rigorous, detailed assessment of the condition of a facility performed to generate a fundable scope and cost estimate for prioritization and funding of maintenance and repair.
Job or Work Order	An authorization for work that requires planning and estimating and has an individual line of accounting for financial and performance evaluation.
Planned Utility Outage	A Planned Utility Outage is a scheduled utility service interruption at or before the utility service point affecting at least one occupied facility, pier, or berth. The Service/Trouble Desk personnel gather pertinent information from customers prior to the outage event and from the field after the event, and then they record the outage event in the Maximo™ Work Order Outage Log. These events require planning and coordination with customers at a minimum of 10 days for routine (and 3 to 4 days for urgent requests) in advance of the outage. Planned utility outages that occur without notification to proceed (after a request to defer the outage cannot be met) are reclassified as unplanned utility outages. Deferred outages are reserved for critical facilities that would sustain severe mission impact if an outage were to occur.
Reportable Utility Outage	A reportable utility outage is defined as a loss of NAVFAC delivered utility services to an occupied facility, pier or berth that is greater than 5 minutes in duration. Utility system failures that do not cause a disruption of services to occupied facilities, piers, or berths are not considered an outage.

ATTACHMENT J-1601000-01
DEFINITIONS AND ACRONYMS

Definition/Acronym	Description
Supervisory Control And Data Acquisition (SCADA)	A version of telemetry commonly used in wide-area industrial applications, such as electrical power generation and distribution and water distribution, which includes supervisory control of remote stations as well as data acquisition from those stations over a bidirectional communications link.
System Average Interruption Duration Index (SAIDI)	SAIDI is defined as the average duration of interruptions per consumer during the year. It is the ratio of the annual duration of interruptions (sustained) to the number of consumers. If duration is specified in minutes, SAIDI is given as consumer minutes. [SAIDI = Total duration of sustained interruptions in a year / Total number of consumers]. SAIDI is an Installation level metric that is the yearly average outage duration for the average customer served at a particular Installation. The electrical reliability benchmark is 120 minutes per year.
System Average Interruption Frequency Index (SAIFI)	SAIFI is defined as the average number of sustained interruptions per consumer during the year. It is the ratio of the annual number of interruptions to the number of consumers. [SAIFI = Total number of sustained interruptions in a year / Total number of consumers]. SAIFI is an Installation level metric that is the yearly average number of interruption occurrences experienced by a customer served at a particular Installation. The electrical reliability benchmark is two (02) occurrence per year.
Service Point	The service point is the point of connection between the serving utility and the facility and is also the 'Point of Demarcation' where ownership changes. This is the transition between NAVFAC UEM and customer funding for maintenance and repair services on equipment.
Unplanned Utility Outage	An Unplanned Utility Outage is an unscheduled utility service interruption at or before the utility service point affecting at least one occupied facility, pier, or berth, and lasting longer than 5 minutes in duration. Service/Trouble Desk personnel enter information from the field about the incident, response, and the repair in a MAXIMO Work Order Outage Log. Facilities experiencing an unplanned utility outage greater than five minutes should contact their Building Monitor or Facility Coordinator who can report the outage to the Service/Trouble Desk.
Utility Investment Strategy	The Contractor shall engage in the Navy's utility investment strategy to provide efficient and effective utilities that directly support warfighter readiness at the lowest possible life cycle cost, while maintaining fully functional capabilities.
Utilities Maintenance Management	The planning, prioritizing, organizing, controlling, reporting, evaluating, and adjusting of utility maintenance operations to support the CNO/NAVFAC utilities policy and objectives and satisfy customers' facility needs. Defined by the International Facility Management Association as "the practice of coordinating the physical workplace with the people and work of the organization."
Utility Service Interruption	A Momentary Utility Service Interruptions of five minutes or less in duration are not considered a long duration utility outage, and therefore not required to be recorded in the MAXIMO system. NAVFAC Utilities does not provide nor guarantee uninterruptible utility services.
Utilities Infrastructure Condition Assessment Program (UICAP)	A Navy automated data management program that utilizes historical asset lifecycle data and a structured assessment process to evaluate the condition of utility systems and their components.
AC	Alternating Current (electricity)
AIS	Automated Information System
AMI	Advanced Metering Infrastructure
ANSI	American National Standards Institute
ArcGIS	Geographic Information System (developed by ESRI)
BCS	Building Control System
CAIDI	Customer Average Interruption Duration Index
CIRCUITS	Centralized and Integrated Reporting for the Comprehensive Utilities
DI	Degradation Index

ATTACHMENT J-1601000-01
DEFINITIONS AND ACRONYMS

Definition/Acronym	Description
DDC	Direct Digital Control Systems
ESRI	Environmental Systems Research Institute
FPOC	Facility Point of Connection
FRCS	Facility Related Control System
GIG	Global Information Grid
GIS	Geospatial Information System
ICS	Industrial Control Systems
IMP	Integrated Maintenance Program
iNFADS	Internet Navy Facilities Assets Data Store
SAIDI	System Average Interruption Duration Index
SAIFI	System Average Frequency Interruption Index
SCADA	Supervisory Control And Data Acquisition
STIG	Security Technical Implementation Guide
WIB	Work Induction Board
WIS	Work Induction System
UEM	Utilities and Energy Management
UEM-DI	Utilities and Energy Management Degradation Index
UICAP	Utility Infrastructure Condition Assessment
COF	Consequence of Failure
LOF	Likelihood of Failure
UEM	Utility Energy Management
UFC	Unified Facilities Criteria
UMIG	Utilities Mission Integration Group
UCS	Utility Control System

ATTACHMENT 1601000-02
REFERENCES AND TECHNICAL DOCUMENTS

Reference	Title
UFC 3-540-01	Engine-Driven Generator Systems For Backup Power Applications
UFC 3-230-01	Water Storage, Distribution, And Transmission
UFC 3-501-01	Electrical Engineering
UFC 3-550-01	Exterior Electrical Power Distribution
MO-221	Utilities Metering
MO-321	Facilities Management
MO-209	Engineering Criteria
UFC 3-501-01	Electrical Engineering
UFC 3-550-01	Exterior Electrical Power Distribution
UFC 3-560-01	Electrical Safety O&M
UFC 4-010-06	Cybersecurity FRCS
SECNAV M-5239.3	Department of the Navy Cybersecurity Manual
AR 420-49	Heating, Energy Selection and Fuel Storage, Distribution, and Dispensing Systems
NAVFAC P-602	3 Pillars of Energy Security
NAVFAC P-603	Utilities Infrastructure Management Manual

ATTACHMENT J-1601000-03
ANNUAL HISTORICAL UTILITY SERVICE ORDER WORKLOAD

Service Order Type	Utility Annex		
	Electrical	Wastewater	Water
Emergency	120	278	66
Routine	1,062	844	130

ATTACHMENT J-1601000-04
UTILITY MAP ATTRIBUTES

Refer to MS Excel file “J-1601000-05 Utility Map Attributes Management.xlsx.” for information.

ATTACHMENT J-1601000-05
UTILITY OUTAGE REPORTING REQUIREMENTS

In accordance with the BPMS B-5.2.14 and NAVFAC MAXIMO data requirements provided in Spec Item 2.6.8 in Annex 0200000, the Contractor shall log and track all utility planned and unplanned outages including the following information for work order data related to utility outages:

- Receive service call.
- Create Service Request for the utility outage following the Service Desk Guide for Utility Outages.
- Create Work Order for response by utilities personnel as soon as the outage is reported and/or when the service request is received. Initial Work Order shall be recorded in NAVFAC MAXIMO no later than the close of next business day.
- Utility outage responding personnel will report the number of buildings and pier berths affected by the outage to the Service/Trouble Desk.
- Update outage information and actions taken to restore service on Work Order as information becomes available to include: for example, Master System Affected, Outage Start Date, Number of Buildings Affected, Outage End Date, and outage cause. Once the work is completed, update records to add any missing data within 30 days and prior to work order close-out.

Utility Outage Cause Codes:

Code 0 - False Alarm - A False Alarm, determined not an outage

Code 1 - Commercial Power/Utility Loss - Outage caused when commercial service provider fails to supply a NAVFAC utility distribution system, such as a loss of generation

Code 2 - Weather - Outage caused by named storms, tornados, lightning strikes, flood, or during a severe weather warning / watch

Code 3 - Equipment Failure - Outage caused by failed circuits, transformers, breaks, leaks, etc.

Code 4 - Animal - Outage caused by animal/pest damage from birds, squirrels, reptiles, etc.

Code 5 - Vegetation - Outage caused by vegetation damage not due to severe weather, such as those due to neglecting to maintain of right of ways

Code 6 - Foreign Interference - Outage caused by excavation, customer interference, vandalism, etc.

Code 7 - Ship - Outage caused by actions aboard ship or by ship personnel

Code 8 - Operational Error - Unplanned outage caused by utilities personnel

Code 9 - Navy Power/Utility Loss – Outage by other utility causes, such as planned.

ATTACHMENT J-1601000-06
UTILITY ASSET TAGGING GUIDANCE

The Contractor shall label all utility assets following the format below. All fields indicated align to asset data stored in NAVFAC MAXIMO.



Installation of these tags shall be phased in over the base year to occur as inventory is completed. As equipment is replaced or new equipment installed, the Contractor shall attach new tags and provide an update of all equipment information within the NAVFAC Maximo. The Contractor shall regularly maintain and replace tags found to be damaged, deteriorated, or unreadable. The application for printing barcode labels and instructions is also available in P-603.

All equipment tags used shall be synthetic, durable, resist caustics/acids and abrasion, and shall be water, heat and weather resistant. Tags shall be made of polypropylene, thermal transfer printable labels with a matte finish, with thickness of 2.4 mil. Affixing method with a permanent-all temp adhesive. The dimensions are 2.43 inches by 3 inches to fit into the Zebra Technology International P4T Mobile Thermal Transfer Printer with ink ribbon or similar.

Equipment tags shall match the asset number of the equipment per record in NAVFAC Maximo.

ATTACHMENT J-1601000-07
SPECIFICATION ATTRIBUTES

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
Actuator	Actuator Type
Actuator	Rated Pressure
Actuator	Rated Pressure UOM
AirDryer	Air Dryer Type
Barscreen	Bar Screen Type
BatterySystem	Battery Role
BatterySystem	Battery Type/Battery Ampere Hour
BatterySystem	Number Of Batteries
Boiler	Type ID
Boiler	National Board No.
Boiler	Capacity
Boiler	Capacity UOM
Boiler	Primary Fuel Type
Boiler	Secondary Fuel Type
Burner	UL Number
Burner	Fuel Type
CapacitorBank	Cap Bank Type
CapacitorBank	Control Type
CapacitorBank	Install Type
CapacitorBank	Nominal Voltage
CapacitorBank	Voltage Class
CapacitorBank	Total Bank KVAR
ChemicalAddition	Chemical Type
Chiller	Capacity
Chiller	Capacity UOM
Chiller	Compressor Type
Compressor	NFAID
Compressor	Compressor Type
CondensateCollector	Steam Condensate Collector Type
CondensateCollector	Associated Line Type
ControlSystem	Type
ControlSystem	Voltage
ControlValve	Type
ControlValve	Diameter
ControlValve	Diameter UOM
ControlValve	Regulation Type
Classification	Required Specification Field
ControlValve	Material
DeaeratorAndStorageTank	Deaerator Type
DeaeratorAndStorageTank	Deaerator Capacity
DeaeratorAndStorageTank	Deaerator Capacity UOM

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
DeaeratorAndStorageTank	Storage Tank Volume
DeaeratorAndStorageTank	Volume UOM
DeaeratorAndStorageTank	Tank Material
DifferentialRelay	Differential Relay Type
DifferentialRelay	Calibration Date
DynamicProtectiveDevice	Dyn Pro Dev Type
DynamicProtectiveDevice	Device Control
DynamicProtectiveDevice	Voltage Class
DynamicProtectiveDevice	Interr Mechanism
DynamicProtectiveDevice	Gang Operated
DynamicProtectiveDevice	Interr Medium
DynamicProtectiveDevice	Install Type
DynamicProtectiveDevice	Nominal Voltage
ElectricLine	Electric Line Type
ElectricLine	Circuit Length
ElectricLine	Length UOM
ElectricLine	Voltage Class
ElectricLine	Conductors per Phase
ElectricLine	Install Type
ElectricLine	Nominal Voltage
ElectricLine	Radial Feeder
ElectricLine	Cable in Conduit
ElectricMeter	Electric Meter Role
ElectricMeter	Meter Form
ElectricMeter	Meter Class
ElectricMeter	Nominal Voltage
ElectricPanel	Panel Type
ElectricPanel	Voltage
ElectricPanel	Amp Rating
ElectricPanel	Number Spaces
ElectroMechanicalRelay	Relay Type
ElectroMechanicalRelay	IEEE Device No
ElectroMechanicalRelay	Calibration Date
ElectroMechanicalRelay	Voltage Class
Enclosure	Enclosure Type
Engine	Driven Asset
Engine	Fuel Type
ExpansionJoint	Steam Exp Joint Type
ExpansionJoint	Assoc Line Type
ExpansionJoint	Diameter
ExpansionJoint	Diameter UOM
ExpansionJoint	Material
ExteriorLights	Exterior Lights Type

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
ExteriorLights	Nominal Voltage
ExteriorLights	Light Ownership
ExteriorLights	Wattage
ExteriorLights	Lamp Type
ExteriorLights	Quantity
Fan	Fan Role
Filter	Filter Type
Filter	Filter Material
Filter	Capacity
Filter	Capacity UOM
Filter	Rated Flow
Filter	Rated Flow UOM
GearDrive	Shaft Alignment
GearDrive	Rated Power
GearDrive	Power UOM
Generator	Rated KW
Generator	Generator Type
Generator	Voltage Class
Generator	Nominal Voltage
GreaseTrapAndOilWaterSeparator	Pretreatment Device Type
GreaseTrapAndOilWaterSeparator	System Type
GreaseTrapAndOilWaterSeparator	Material
GritRemoval	Grit Removal Type
Ground	Grounding Device Type
Ground	Ground Type
HeatExchanger	Heat Exchanger Role
Hydrant	Year Manufactured
Hydrant	Bonnet Color
Hydrant	Material
Hydrant	Hydrant Type
Hydrant	Seat Diameter
Hydrant	Diameter UOM
Instrumentation	Instrumentation Type
Insulator	Insulator Type
Insulator	Insulator Material
Insulator	Install Type
Insulator	Voltage Class
IonExchangeUnit	Capacity
IonExchangeUnit	Capacity UOM
LVBreakerPanel	Install Type
LVBreakerPanel	Nominal Voltage
LVBreakerPanel	Wire
LVBreakerPanel	Rated Ampacity

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
LVBreakerPanel	Number Phases
LVBreakerPanel	Number Spaces
LVBreakerPanel	Spaces Filled
MechanicalMeter	Mech Meter Role
MechanicalMeter	Mech Meter Billing Type
MechanicalMeter	Remote Read Indicator
MechanicalMeter	Material
MechanicalMeter	Meter Type
Mixer	Mixer Type
Motor	Motor Type
Motor	Motor Control Type
Motor	RPM
Motor	Rated HP
Motor	Nominal Voltage
Motor	Voltage Class
OvercurrentRelay	Overcurrent Relay Type
OvercurrentRelay	Calibration Date/Setting of Relay
Pipe	Service Area
Pipe	Line Type
Pipe	Pipe Length
Pipe	Pipe Length UOM
Pipe	Material
Pipe	Pipe GPM
PowerInverter	Nom. Input Volt.
PowerInverter	Nom. AC Volt. Out
PowerInverter	Cont/Nom Pwr Out./Ampere
Pump	Type
Pump	Submersible Indicator
Pump	Rated Flow
Pump	Rated Flow UOM
Pump	Mechanical System Type
Pump	Material
Pump	Electrical Power Demand
Receptacle	Receptacle Type
Receptacle	Install Type
Receptacle	Voltage Class
Receptacle	Nominal Voltage
Receptacle	Rated Ampacity
ReliefValve	Type
ReliefValve	Diameter
ReliefValve	Diameter UOM
ReliefValve	Material
RotatingBiologicalContactor	Capacity

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
RotatingBiologicalContactor	Capacity UOM
ShorePowerBox	Shore Power Box Type
ShorePowerBox	Voltage Class
ShorePowerBox	Nominal Voltage
ShorePowerBox	Install Type
SolidStateRelay	Relay Type
SolidStateRelay	IEEE Device No
SolidStateRelay	Calibration Date
SolidStateRelay	Voltage Class
Stack	Material
Stack	Height
Stack	Height UOM
Stack	Mult. Boilers
Stack	Mounting System
StaticProtectiveDevice	Device Type
StaticProtectiveDevice	Install Type
StaticProtectiveDevice	Voltage Class
SupportStructure	Structure Material
SupportStructure	Supp Struct Type
SupportStructure	Thermal Support Type
SurfaceStructure	Structure Material
SurfaceStructure	Surface Structure Type
SwitchUnit	Switch Type
SwitchUnit	Voltage Class
SwitchUnit	Sw Insul Medium
SwitchUnit	Nominal Voltage
SwitchUnit	Gang Operated
SwitchUnit	Install Type
SystemValve	Type
SystemValve	Diameter
SystemValve	Diameter UOM
SystemValve	Material
Tank	Type
Tank	Material
Tank	Contents
Tank	Volume
Tank	Volume UOM
Transformer	TransformerType
Transformer	Rated KVA
Transformer	Install Type
Transformer	Nominal HV
Transformer	HV Protection
Transformer	Nominal LV

CLASSIFICATION	REQUIRED SPECIFICATION FIELD
Transformer	CoolingType
Transformer	HSide Taps
Trap	Type
TreatmentProcessStructure	Capacity
TreatmentProcessStructure	Capacity UOM
UndergroundStructure	Underground Struc Type
UndergroundStructure	Structure Material
VoltageRegulator	Volt Reg Type
VoltageRegulator	Install Type
VoltageRegulator	Rated KVA
VoltageRegulator	Nominal Voltage
Well	Rated Flow
Well	Rated Flow UOM

ATTACHMENT J-1601000-08
UTILITY ASSET INVENTORY

Refer to MS Excel file “J-1601000-09 Utility Asset Inventory.xlsx.” for information.

ATTACHMENT J-1601000-09
UEM-DI METER GROUP AND THE DIRECT CONDITION RATING GUIDANCE

	Likelihood Category	Degradation Index
Fully Functional	Negligible = 0	No noticeable defects. Some aging or wear may be visible. Fully functional.
	Not Likely = 1	Only minor deterioration or defects are evident. Noticeable wear or aging is visible. Fully functional. Minor maintenance may be required.
	Possible = 2	Deterioration or defects are evident. Function is not significantly affected. Minor repairs are required.
Function Affected	Likely = 3	Serious deterioration or defects in at least some portion of the asset. Function is significantly affected. Extensive repairs or replacement are required.
	Very Likely = 4	Extensive deterioration or defects in the asset. Not functional, barely functional, or beyond repair.